Submit your source code file as well as a brief report with the following components:

- for each epsilon value in {0.01, 0.1, 1, 10}, report the MAE for your differentially private histogram

0.01 - 294.4189189189189

0.1 - 37.898648648648646

1 - 3.7905405405405403

10 - 0.30405405405405406

- discuss the above observations and what you learn from this implementation exercise

Creating this program was very similar to HW3 just adding the noise was a little different. It helped me learn how noise is added during Gaussians mechanism and showed that laplace would be better for data like this.

- In addition, contrast the results you obtained this time with the results of Laplace mechanism in HW3, and provide your own interpretations on any differences.

Looking back at my laplace mechanism for the same data, I get a lower mae for each epsilon. More noise is added during the gaussian mechanism because it has the small failure probability of delta added to it. For this data the laplace mechanism is better because it has a smaller mae for each epsilon value and will still result in more privacy then gaussian.